

Raven's Test of Intelligence

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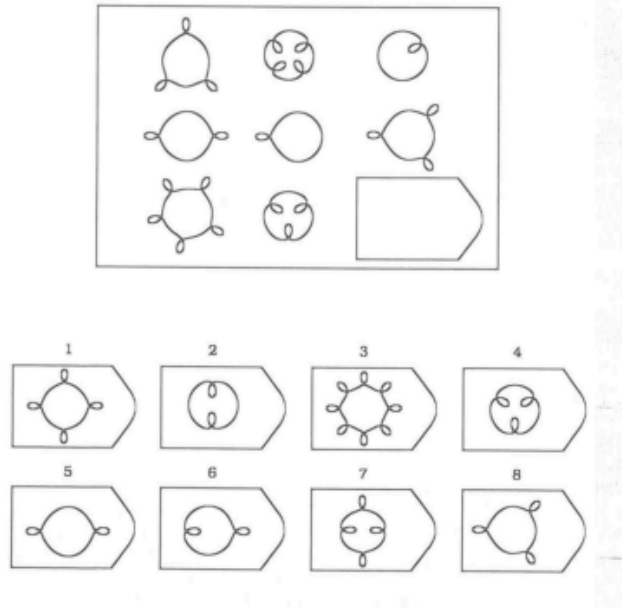


Fig. 1. Raven's Test of Intelligence

I. RAVEN'S TEST OF INTELLIGENCE

Based on the intelligence problem, my concluded answer to the Raven's test of intelligence would be option 5. To arrive at this conclusion I first separated by thought process into two sections, determining how many loops the image would have and whether or not the loops would be outside or inside. To identify how many loops on the outside we must look at the column of each matrix. The first column as outside loops of 3, 2, and 5 respectively. Two of the number of loops in the image combine to form the third. This pattern continues to follow for the second column of 1 outside loop and 3 and 4 inside loops. If we identify the inside loops as negatives then $1 + (-3) = -4$ which also satisfies the second column. This means that the third column would be 2 outside loops, since there is a 3 outside loop and 1 inside loop, so $-1 + 3 = 2$. We can confirm that 2 outside loops would be the correct answer by confirming the matrix works horizontally as well. The formula would be to add the first and second column to get the value of the third column. The first row would be $3 + (-4) = -1$. The second row would be $2 + 1 = 3$, and the last row would be $5 + (-3) = 2$. This confirms that option 5 which is the image with 2 outside loops is the correct answer.

II. REPRESENTATION

A. Analogies

In this specific Raven's Test of Intelligence, analogies are definitively engaged since the matrix itself forces you to work with the images as rows and columns and not just individual images. You compare relationships across row and columns and then apply that same relationship to the missing space to identify the answer. For example, the first row has 3 outside loops then 4 inside loops and then 1 inside loop. The second row has 2 outside loops then 1 outside loop and then 3 outside loop. By analogy, the third row would add and subtract loops just like the first two rows.

B. Imagistic Representations

Just like analogies, imagistic representations are required for solving this test. Imagistic representations can be through the loops outside and inside the circle. Imagistic reasoning uses visual representations to work through the reasoning behind a problem and in this test you can visually see the inner loops and outside loops changing from row to row and column to column [1]. For example, in the middle row you can physically see the first image having 2 outside loops. Then by growing the singular loop on the middle column, the circle becomes three outside loops like the third column.

C. Conceptual Representations

Personally I believe that this problem is best requires conceptual representations because I solved the problem easiest through this method. Conceptual representation in this test involves using symbolic representation of the images to transition them into mathematical problems that can then be solved using logical rules [2]. If you take out all the loops and circles of the image and swap them out for positive and negative numbers, where the outside loop are positive numbers and the inside loop is negative numbers, then solving the problem itself becomes very simple. The conceptual representation of the loops becomes numbers and the rules behind the matrix becomes clear. For example, the first row with 3 outside loops, 4 inside loops, and 1 inside loops becomes 3, -4, and -1 respectively, allowing you to see the pattern of adding the numbers together.

III. SOLUTION APPROACH

Each of the representations build upon each other to get you to the correct answer. However, I believe my approach is most heavily dependent on logic and rule based reasoning. Even

though both analogous thinking and imagistic representation were used, they were more supplemental in the solution compared to the main solution that was ultimately concluded through logic. I came to this conclusion because logic was still the main process in which the problem was solved. The test still came down to being able to mathematically identify the missing piece of the puzzle. Imagistic and analogical reasoning was still present in the process, such as visually comparing the outside loops and inside loops and the pattern inferences of the rows and column, but a majority of the test revolved around the logical reasoning of the test. Once the loops and analogy of the rows and columns were complete the matrix itself was a completely mathematical problem that required logic based processing to complete. Overall, my solution approach could be said to encompass all three types of representation, but more heavily leaning towards logic based reasoning methods.

REFERENCES

- [1] "Mental Images: In search of a Theory," *Ucsd.edu*, 2025. <https://pages.ucsd.edu/~scoulson/203/pylyshyn.html> (accessed Sep. 08, 2025).
- [2] D. Pitt, "Mental Representation (Stanford Encyclopedia of Philosophy)," *Stanford.edu*, 2012. <https://plato.stanford.edu/entries/mental-representation/>